## PUBLIC SOURCES

### World of Logs v2

public Jira issues, each flagged for whether its text contains log output

### Public Jira Dataset

creation and resolution dates, and duplicate links between issues

### Four git histories

Apache Kafka (Java)
MariaDB (C / C++)
MongoDB (C++)
Qt (C++)

*four organizations, two tracker platforms*

## MINE THE BENCHMARK

### Join, then filter

keep fixed Bug issues whose text carries logs, link them to 1-3 fix commits by issue key

| | |
|---|---|
| eligible issues | 17,995 |
| no linkable fix commit | -9,134 |
| no source file in the fix | -2,206 |

### Ground truth, from git

the snapshot is the parent of the fix commit; the gold files are the source files it modified

*never a file the fix only added*

### Incident memory pool

every resolved issue with a linked fix, kept per project

*2,315 to 19,820 incidents*

## THE BENCHMARK

**6,437**
**real incidents**

each one pairs a report and its logs, the code of that day, and the files the fix later changed

### Stratified by difficulty

in **53%** of reports, no file the fix touched is named anywhere

**Two evaluation populations**

**Full evaluation split 4,508**

later 70% of each project by fix time; benchmark-wide measurements

**Study population 790**

200 per project, test-only gold removed; **every** system rescored here

## COMPARE, ON ONE POPULATION

**one protocol, one temporal firewall**

*Acc@5*

| | |
|---|---|
| **Classic IR localization**<br>rVSM, BugLocator, BRTracer-style | 18.0% |
| **Retrieval + incident memory**<br>lexical, deterministic, CPU only | 68.1% |
| **A 27B model, given the file tree**<br>Agentless-style; re-ranker variant 78.0% | 83.2% |
| **Fusion of the two above**<br>best system that is not an agent | 83.9% |
| **REMEMBER (this paper)**<br>same 27B model, given tools | 90.3% |

No family dominates. Retrieval and the model solve **different** instances; union 89.2%

## CHECK THE RESULT

### Did it memorise?

asked for each fix from the issue key alone, the model scores **below** a frequency prior

### Is it echoing retrieval?

shuffling the candidate order changes **no** result, while its own picks overlap drops 60% to 21%

### Is the memory honest?

dropping the temporal firewall inflates the classic similar-report signal by **9% to 43%**

### Every number

traces to a result file through a released script, check by check